GigBid: a multi-agent freelancer copilot

AI-4016 R&D report · Hasan · FAST-NUCES Islamabad · repository commit `a1de0db` · results profile `heuristic` · providers actually used in the main run: {'heuristic': 600}

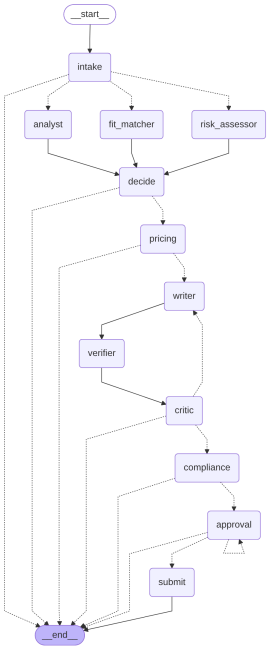
> **Read first: threats to validity.** (1) The instructor GigSim, persona corpora and dev set were not available; every experiment uses this repository's GigSim stand-in (`gigsim/`, seeded and reproducible), written by the same author as the system. (2) The workspace had no LLM API access while these numbers were produced, so results come from the `heuristic` profile, the deterministic rule-based fallback every node has (ADR 0004). Experiments that only make sense with a model (tool-granularity study, single-agent comparison, judge calibration) have committed runners and are marked pending. (3) Simulated-client reply rate is a designed proxy. (4) Real Upwork data (six Kaggle datasets, §9) arrived late; it has no gold labels, so it is used for a real job board, real prices and checks of the rules, while the labelled metrics stay on the synthetic sets.

1. Problem and constraints

Freelancers spend hours scanning feeds, spend paid Connects on jobs they won't win, send generic proposals, and run into scams that ask for free work or off-platform payment. GigBid automates discovery, assessment and drafting. The freelancer makes every decision that leaves the system.

- **Never automated:** submitting a proposal, sending a message, spending Connects, claiming experience the portfolio doesn't prove, contacting a client off-platform. Enforced by a LangGraph `interrupt()` *and* an HMAC approval token checked inside the Marketplace MCP server (ADR 0002).
- **Allowed automatically:** scouting, analysis, skipping (only with autonomy `auto_skip_low_fit`), drafting.
- **Budgets** (config, enforced per thread): 40 graph steps, 30k tokens per job, max 2 Critic revisions; latency and cost budgets in §11.

2. Architecture



Bid graph (one thread per job): intake → `Send` to Job Analyst ‖ Risk Assessor ‖ Fit Matcher → decide (bid/watch/skip with reasons) → Pricing (can veto) → Writer → claim Verifier → Critic (≤2 loops) → Compliance → approval interrupt → submit (non-LLM executor, idempotency keys derived from the thread).
Conversation graph (one thread per client conversation): load → classify intent (+injection) → calendar slots → draft → compliance → approval → send + tentative hold.

Marketplace port (`agent/marketplace_port.py`) has two adapters, GigSim and live Upwork (official MCP over Streamable HTTP, OAuth 2.1). The port lives inside our Marketplace MCP server, so the graph cannot tell which marketplace it uses.
MCP surface: Portfolio (read, publishable), Marketplace (read + write with schemas, annotations, idempotency keys, `dry_run`, bearer auth, per-user limiter, hash-chained audit log), Workspace (calendar, email drafts), plus GitHub MCP for repo ingestion. Each agent gets an allowlist of tools. **Stores:** Postgres (app tables, LangGraph checkpointer, LangGraph store for memory), Redis (Arq queue), in-process vectors + BM25 (ADR 0003).

Why each agent exists: parallel analysts cut wall-clock time and keep the Analyst tool-less (an injected post can't trigger tool calls). Fit Matcher makes "no evidence" explicit per requirement. Pricing computes numbers in code and the model only explains them. Verifier, Critic and Compliance are separated because each failure needs a different fix (§8). Typed hand-offs: `JobSpec, RiskReport, FitReport, BidDecision, PriceQuote, ProposalDraft, Critique, ComplianceVerdict`. State, execution trace and long-term memory are kept apart (docs/ARCHITECTURE.md).

Durability (C6) is tested in `tests/integration`: the graph pauses at approval, the process is replaced, approval resumes from the Postgres checkpoint, and replaying the `submit` node from the previous checkpoint returns `idempotent_replay` with exactly one GigSim submission and one Connects deduction. Public traces: `docs/traces/` (scam skip, approval, injection, Critic loop, crash-and-resume).

3. R&D question A: tool granularity

Hypothesis: a small set of mid-grained tools with rewritten, disambiguating descriptions beats both 17 fine-grained tools (selection errors between near-duplicates) and 3 coarse tools with an `action` enum (argument errors) on tool-selection and

argument accuracy, at fewer tokens per task than fine-grained. **Setup:** 40 tasks with known correct calls (`evals/datasets/tool_tasks.jsonl`), same model, native tool calling, `tool_choice=required`. **Results:**

Tool set	# tools	Tool selection acc.	Argument acc.	Tokens / task
fine	17	1.0	0.9737	719.6316
coarse	3	0.7895	0.1316	345.1316
best	10	1.0	1.0	543.3947

[tool_study.json]

Shipped design: 10 read tools + 3 write tools; descriptions say what the tool is *not* for ("never for details of one known job id"). Contract tests pin schemas and annotations (`tests/contract/snapshots`).

4. R&D question B: RAG variants, claim verification, shipped choice

Hypothesis (stated before running): requirement decomposition plus hybrid retrieval raises evidence Recall@5 by 5+ points over V0 on jobs with 4+ requirements. **Result: rejected.** Alone, V1 and V2 lowered Recall@5. Corpus: 384 dev jobs with gold evidence; retrieval unit = project.

Variant	Recall@5	MRR	nDCG@10	R@5 (4+ reqs)	note
v0	0.7583	0.9445	0.8333	0.7317	
v1	0.7488	0.9546	0.8357	0.7222	
v2	0.7296	0.9494	0.8378	0.6999	
v3	–	–	–	–	ModuleNotFoundError: No module named 'sentenc
v4	0.7845	0.974	0.8071	0.7605	
v5	0.7809	0.9754	0.8427	0.7489	
v6	0.7616	0.9732	0.8492	0.7378	
v7	0.7395	0.892	0.7936	0.7181	
v8	0.7779	0.9824	0.8208	0.741	
best	0.7662	0.9894	0.8799	0.7298	
best-no_decompose	0.7684	0.986	0.8682	0.7415	
best-no_bm25	0.8041	0.9912	0.8774	0.7711	
best+multivector	0.7819	0.9232	0.8172	0.7508	
best-no_context	0.7512	0.9856	0.8636	0.7126	
best-fixed_chunks	0.7338	0.9885	0.8661	0.6902	

[rag_heuristic.json]

A 64-config sweep with nDCG@10 declared as the primary metric before running picked decompose + hybrid + contextual header + structural chunks:

Config (query-retrieval-mv-ctx-chunk-outcome)	nDCG@10	R@5	MRR
dec-hyb-mv0-cx1-str-ob0	0.8799	0.7662	0.9894
dec-den-mv0-cx1-str-ob0	0.8774	0.8041	0.9912
dec-den-mv1-cx1-str-ob0	0.8725	0.8124	0.989
who-hyb-mv0-cx1-str-ob0	0.8682	0.7684	0.986
dec-den-mv1-cx1-fix-ob0	0.8662	0.7866	0.9908
dec-hyb-mv0-cx1-fix-ob0	0.8661	0.7338	0.9885

[rag_sweep_heuristic.json]

Ablation of the shipped pipeline (one component at a time): no decomposition → nDCG 0.8682; no BM25 → 0.8774 (R@5 rises to 0.8041); no contextual header → 0.8636; fixed chunks → 0.8661; adding multi-vector → 0.8172. Context headers and structural chunks carry the gain. Held-out: V0 0.8394 vs shipped 0.8634 nDCG@10. V3 (bge-reranker-v2-m3) could not be run: weights on Hugging Face were unreachable from the build workspace.

End-to-end (results table required by B5)

Variant	Evidence R@5	nDCG@10	Unsupported / proposal (before → after)	Judge	GigSim reply	Hire	p95 s/job	USD/proposal
V0	0.7583	0.8333	0.0466 → 0.0	n/a	0.8347	0.3729	0.3935	0.0
V5	0.7809	0.8427	0.1549 → 0.0	n/a	0.8761	0.4823	0.4915	0.0
V8	0.7779	0.8208	0.0586 → 0.0	n/a	0.8604	0.455	0.4068	0.0
Shipped	0.7662	0.8799	0.0556 → 0.0	n/a	0.8803	0.406	0.4347	0.0

[pipeline_heuristic_dev*.json]

By job type:

Job type	R@5 V0	R@5 shipped	nDCG V0	nDCG shipped
1-3_requirements	0.8011	0.8248	0.8718	0.9133
4+_requirements	0.7317	0.7298	0.8094	0.8593

By persona:

Persona	Bid F1	Scam P	Scam R	Price in band	Proposals	Reply rate	R@5 shipped	R@5 V0
ai_ml	0.9474	1.0	1.0	0.7317	82	0.878	0.7206	0.7455
fullstack	0.9659	1.0	1.0	0.6512	86	0.8953	0.7969	0.7653
data_analyst	0.942	1.0	1.0	0.7273	66	0.8636	0.7802	0.7639

Claim verification (B3). Claims are split by sentence. Each is checked against cited and retrieved evidence by entity (skills incl. ontology aliases, numbers, proper names; a wrong name or number is always unsupported) and adjudicated by a model only when it has no checkable entities. Unsupported claims are removed before the Critic. Unsupported claims per proposal before verification: 0.0556, after: 0.0. **LLM-judge calibration:** *pending: needs a model endpoint (runner committed)*.

Where better retrieval didn't translate into replies, and why. Reply rates across variants (0.8347–0.8761) moved far less than one verifier rule moved them: before the fix in §8 the shipped pipeline had 0.8362 reply rate with 0.4612 claims removed per proposal; after, 0.8803 with 0.0556. Simulated clients reward "must-have evidenced", which saturates once the top 3 items contain one strong match per must-have. Most variants already reach that, so nDCG gains deeper in the ranking don't show up in replies. **Decision:** ship the sweep winner (`rag.variant: best`); revisit with the model profile.

5. R&D question C: single agent vs supervisor vs graph

Setup: 50 dev jobs (all scams and injections in the sample plus normal jobs), (a) one ReAct agent holding all tools, (b) prebuilt `langgraph-supervisor` with researcher/matcher/writer, (c) the hand-built graph. Same model, same tools, same verifier and policy scan applied to outputs. **Results:** *pending: needs a model endpoint (runner committed)*

Expectation to test: the single agent should match the graph on simple in-domain jobs at lower latency and fail on scams and injections (no rule floor, no separate compliance pass), and on claims (no verifier). Whether to ship it: no, because only the graph can enforce the approval interrupt at a fixed point.

6. The change request

The official 2 Nov "platform policy update" text was not available, so the GigSim stand-in models one new **rule** (no external links except GitHub) and one new **cost** (+2 Connects on jobs with 20+ proposals) as `policy_v2.yaml`. Absorbing it: 1 config line (`policy.version: 2`), 0 code lines, 38 s including the full test suite (31 passed). Rules and costs are data read by the Compliance Guard and Pricing Strategist. **Caveat:** The surcharge and regex-rule mechanisms were designed before the change was known, and the change was authored by the same person, so this is a best case. A rule type the engine cannot express (e.g. time-of-day limits, per-client quotas) would need code in agent/nodes/compliance.py::scan or agent/nodes/pricing.py::connects_cost. [change_request.json]

7. Signature extension: X2 proposal experiments

Strategy A (evidence-first) vs B (question-first, short), paired on the same 234 jobs, scored by GigSim's deterministic clients. Reply rate A 0.8803 vs B 0.7607, hire A 0.406 vs B 0.3803. Discordant pairs 46 vs 18, McNemar exact p = 0.0006. B loses because one cited project leaves the second must-have unevidenced. In production, assignment is deterministic by job id (`experiments.proposal_ab`) and the arm is stored on the draft. [ab_test_heuristic.json]

8. Failure analysis

All failed dev jobs were grouped by class ([failure_analysis*.json]); examples are listed in the appendix with trace ids.

Failure class	before fixes	after fixes
injection_reached_writer	18	16
mispriced	66	66
unsupported_claim_removed	117	13
wrong_bid	28	3
wrong_skip	22	22

- **Wrong bid: high-risk new clients** (28 → 3). Client with no verified payment and zero spend scored risk below 0.6. *Fix:* explicit rule `skip_unverified_new_clients` (config). *Worked:* bid F1 0.9026 → 0.9526.
- **Unsupported claim (verifier false positive)** (117 → 13 removed sentences). 94 of 107 removals were the writer's framing opener, misread as a claim because it contained a skill and "my". *Fix:* a claim needs a past-work assertion. *Worked:* reply rate 0.8362 → 0.8803.
- **Compliance miss (over-blocking)**: the phone regex matched ISO slot dates, so offered interview slots were deleted from replies. *Fix:* stricter pattern + regression test. *Worked* (video, §D).
- **Mispriced** (66): the bid is capped at the posted budget while the gold fair band comes from market history above it. *Not fixed:* bidding above a client's stated budget is a product decision, so I chose to keep the cap and report it.

- **Wrong skip** (22): the Pricing Strategist's floor estimate (market p50 / $45 × complexity) exceeds the budget on cheap categories, so it vetoes jobs the gold labels bid on. *Tried:* nothing yet; the next step is to estimate hours from the job spec, not the category median.
- **Injection reached the writer** (16 drafted, 0 attacks succeeded): defence in depth (delimiting, rule-floor flag, compliance canary/contact rules) held, but the rule-based writer never follows instructions, so the *with/without defence* comparison is uninformative on this profile (0.0 vs 0.0). It must be rerun with a model writer.
- **Tool error / lifecycle**: the MCP idempotency store outlived GigSim state between test runs and replayed a stale proposal id. *Fix:* test runtime isolation. *Lesson:* dedup records need the same lifecycle as the thing they dedupe.
- **Generic opening**: 0 flagged by simulated clients after the verifier fix; the Critic rejects drafts whose first 80 characters match the generic list.

9. Real data (Kaggle): what it changed

Selection. Of the six allowed datasets: 60k Upwork posts (Apache 2.0) → the real job board (1295 posts after cleaning, de-duplication, contact-detail removal and category mapping); 1.3M contracts (ODC-By) → real price percentiles (479,100 hourly, 427,368 completed fixed-price); 2024 posts (50k + tracker, ODC-By, 733,655 unique) → recent budgets, weekly demand and an untouched test sample; the freelancer-success set was **rejected** (third-party synthetic; 0.6266 of rates sit at a $280 cap); the fake-job set (no stated licence) is used only as an outside test, nothing committed. Every job, client and market number carries `source: real|synthetic`; simulated client preferences stay synthetic for everyone. Category mapping, hand-checked on 60 posts: 0.7667 category, 0.85 persona-level. Details, attribution and limits: `docs/DATA_CARD.md`.

Calibration. The synthetic generator's hourly medians vs real contract medians: llm_apps 55 vs 32, web_app 40 vs 25, backend 42 vs 25, bi 32 vs 40, web_analytics 28 vs 46.8, research 28 vs 40. The synthetic world over-prices developer work by 1.4–1.7× and under-prices analyst work. It was not regenerated (that would invalidate the labelled results); real posts are priced from the real contracts instead (hourly rate inside the posted range, or real fixed-price percentiles).

Scam rules v1 → v2.

	v1	v2
Synthetic scam jobs (text rules), precision / recall	0.89 / 1.0	1.0 / 1.0
Real board posts flagged (in-sample)	0.0301	0.0008
3,000 held-out 2024 posts flagged	0.023	0.0017
Dev pipeline scam precision / bid F1	0.9221 / 0.9482	1.0 / 0.9526

v1 fired on any mention of Telegram, WhatsApp or crypto; on real posts these are usually the product ("build a Telegram bot"). v2 needs an ask ("contact me on…", "we pay in USDT"). Of the 5 held-out v2 flags, 1 is a true request for free work, 1 ambiguous, 3 false alarms; v2 misses one "USDT seller" post that v1 caught by accident. [real_scam_external.json]

Outside scam test. On the fake-job set's test half without the template fakes (17,105 ads), the freelance rules catch 0.003 of fakes; TF-IDF + logistic regression trained on its train half gets F1 0.9199 there but recall 0.0 on freelance scams. Job-ad fraud and marketplace scams look different, so the classifier is not shipped.

Real posts through the pipeline (60 posts, no labels): bid rate 0.2889 in the persona's market vs 0.0 outside it; bids inside real p10–p90 1.0, within posted budget 1.0. The main in-market skip reasons are real-market facts the synthetic world under-represents: unverified clients with no spend (30% of real clients are unverified) and budgets below the personas' floors. [pipeline_heuristic_real.json]

10. Safety and platform respect

- **Never:** send or spend without approval (interrupt + HMAC token per content hash), scrape or automate a browser (live mode via the official MCP only, read polling ≥ 240 min, RUNBOOK), share contact details, claim unproven experience, follow instructions in job posts or client messages.
- **How:** untrusted text is delimited and closing tags are neutralised; the Risk Assessor's rule floor can't be lowered by a model; tool allowlists (the Analyst has none, LLM agents have no write tools); Compliance policy pack; PII masking (email, phone, card, IBAN) in logs, traces and the audit log; OAuth tokens Fernet-encrypted; every query scoped by user id.
- **What still gets through:** paraphrased injections without trigger words reach the writer as data (the model must ignore them); portfolio uploads are shared between users who pick the same demo persona (known tenant-isolation gap for demo personas only); simulated clients cannot test whether a real client is misled by a technically-true claim.

11. Hidden-test results against declared budgets

The instructor's hidden run happens after the deadline. The table reports this repository's own held-out set (180 jobs, separate seed, never used for tuning): bid F1 0.9559, scam recall 1.0, precision 1.0, price in band 0.7077, reply rate 0.8769, injection success 0.0. [pipeline_heuristic_heldout.json]

Budget	Declared (agent/config.yaml)	Measured (heuristic profile)
p95 latency per job analysed	20 s	0.4347 s (dev) / 0.4415 s (held-out)
p95 latency per proposal	45 s	0.481 s / 0.5271 s
USD per job analysed	$0.002	$0.0 / $0.0
USD per proposal	$0.01	$0.0 / $0.0

Budgets are met trivially on the rule profile (no model calls). One optimisation that helped: passing the scouted job into the graph instead of re-fetching it, plus running sync MCP tools off the event loop, cut a 15-job scout+analyse batch from about 75 s to under 25 s under GigSim's rate limit. The model-profile numbers will replace these when an endpoint is available.

Appendix A: Reproduce

`make seed` · start GigSim with held-out jobs · `MODEL_PROFILE=heuristic ./scripts/run_all_evals.sh` · `python scripts/update_readme.py heuristic` · `python docs/report/build_report.py heuristic`. Tests: `make test` (32 Python tests), `make e2e` (5 Playwright tests).

Appendix B: failure examples (trace ids)

Class	Job	Category	Action	Gold bid	Price	Fair band	Trace
mispriced	dev-ai-0004	llm_apps	bid	True	2740.0	[3480, 5110]	tr-f258067fc6a543f1
mispriced	dev-ai-0010	nlp	bid	True	2480.0	[1600, 2390]	tr-68da89f915b74402
mispriced	dev-ai-0030	llm_apps	bid	True	3260.0	[3590, 5830]	tr-a859c72070d242c7
wrong_skip	dev-ai-0012	classical_ml	skip	True	1110.0	[1630, 2430]	tr-f345784c2131427d
wrong_skip	dev-ai-0044	mlops	skip	True	2010.0	[2860, 4080]	tr-45193ba218214fee
wrong_skip	dev-ai-0045	vision	skip	True	1900.0	[2350, 3440]	tr-3353e4905c804e48
unsupported_claim_removed	dev-ai-0034	llm_apps	bid	True	4480.0	[3580, 5620]	tr-dc70c9c1cb4e4ce8
unsupported_claim_removed	dev-ai-0039	llm_apps	bid	True	4480.0	[3580, 5620]	tr-de48e597d8854f15
unsupported_claim_removed	dev-ai-0057	llm_apps	bid	True	3050.0	[3580, 5620]	tr-075a9f1ee131448b
injection_reached_writer	dev-ai-0039	llm_apps	bid	True	4480.0	[3580, 5620]	tr-de48e597d8854f15
injection_reached_writer	dev-ai-0043	llm_apps	bid	True	4400.0	[3580, 5620]	tr-3202c8f62dd04f8e
injection_reached_writer	dev-ai-0168	classical_ml	bid	True	1930.0	[1630, 2430]	tr-461e0fe622b64e16
wrong_bid	dev-ai-0196	forecasting	bid	False	660.0	[900, 1410]	tr-c3ab3f006d6740f6
wrong_bid	dev-fu-0080	ecommerce	bid	False	750.0	[1240, 2000]	tr-bec130daa7114304
wrong_bid	dev-da-0199	research	bid	False	350.0	[460, 810]	tr-97a558ab21cb4a4c

Appendix C: AI usage

See `docs/AI_USAGE.md`: tool used, files, and five mistakes caught.